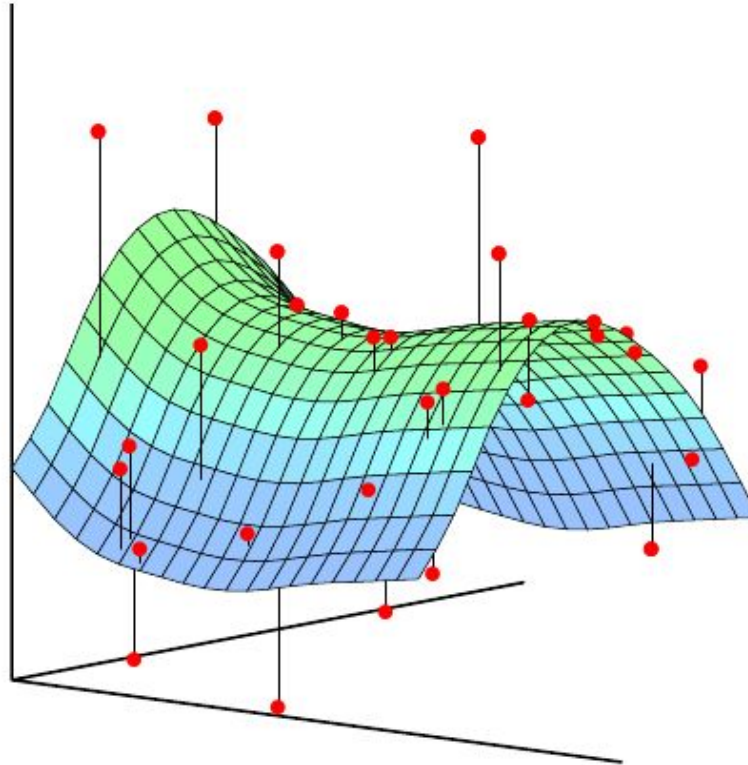


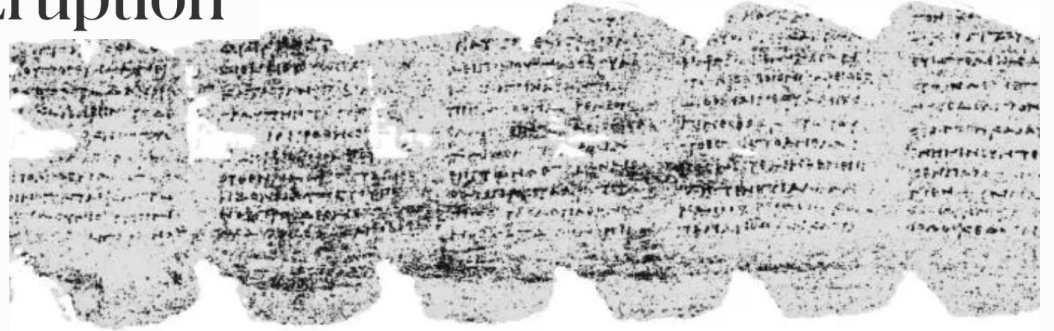
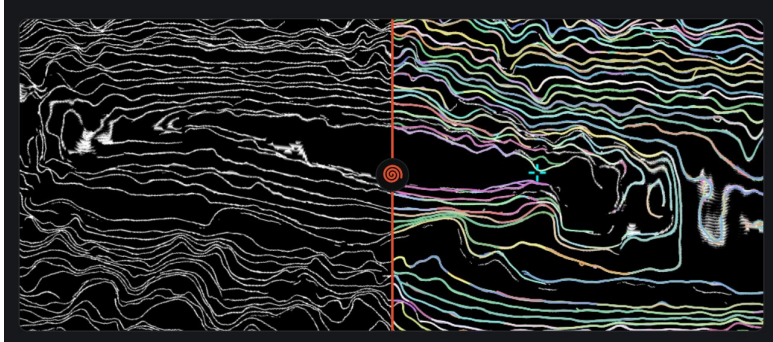
Statistical Learning



Adapted from the *Introduction to Statistical Learning* lecture slides by Trevor Hastie and Robert Tibshirani (statlearning.com), © the authors, used and modified with permission for non-profit educational use. Modifications and any resulting errors are my own.

AI In the News

In A First, AI Decodes Entire Scroll Scorched In Mt. Vesuvius Eruption



The unwrapped surface of the scroll, where columns of Greek writing emerge as the ink signal is recovered.
VESUVIUS CHALLENGE

Lethal snake venom may be countered by new AI-designed proteins

An artificial intelligence tool designs proteins that match toxins scientists want to target



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NEWS AND VIEWS | 18 February 2026

AI succeeds in diagnosing rare diseases

An artificial-intelligence system uses clinical data, genetic information and literature searches to suggest diagnoses and provides the underlying reasoning.

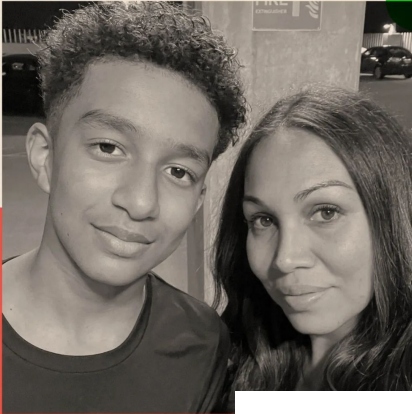
AI In the News

'A predator in your home': Mothers say chatbots encouraged their sons to kill themselves

8 November 2025

 **Laura Kuenssberg**
Presenter, Sunday with Laura Kuenssberg

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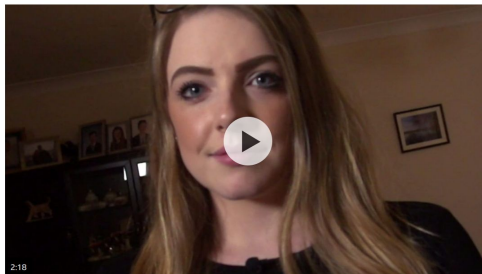
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Instagram eating disorder content 'out of control'

20 March 2019

Angus Crawford
BBC News

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2:18

'My anorexia really loved Instagram'

Content on Instagram that encourages eating disorders is "spiralling out of control", psychiatrists have warned.

NEWS ANALYSIS

A.I. Could Possibly End Humanity. How Are Humans Supposed to Process That?

The technology is far from the only existential risk our species faces. But weighing such cosmic fears can become a mind-bending exercise.

Listen · 8:04 min

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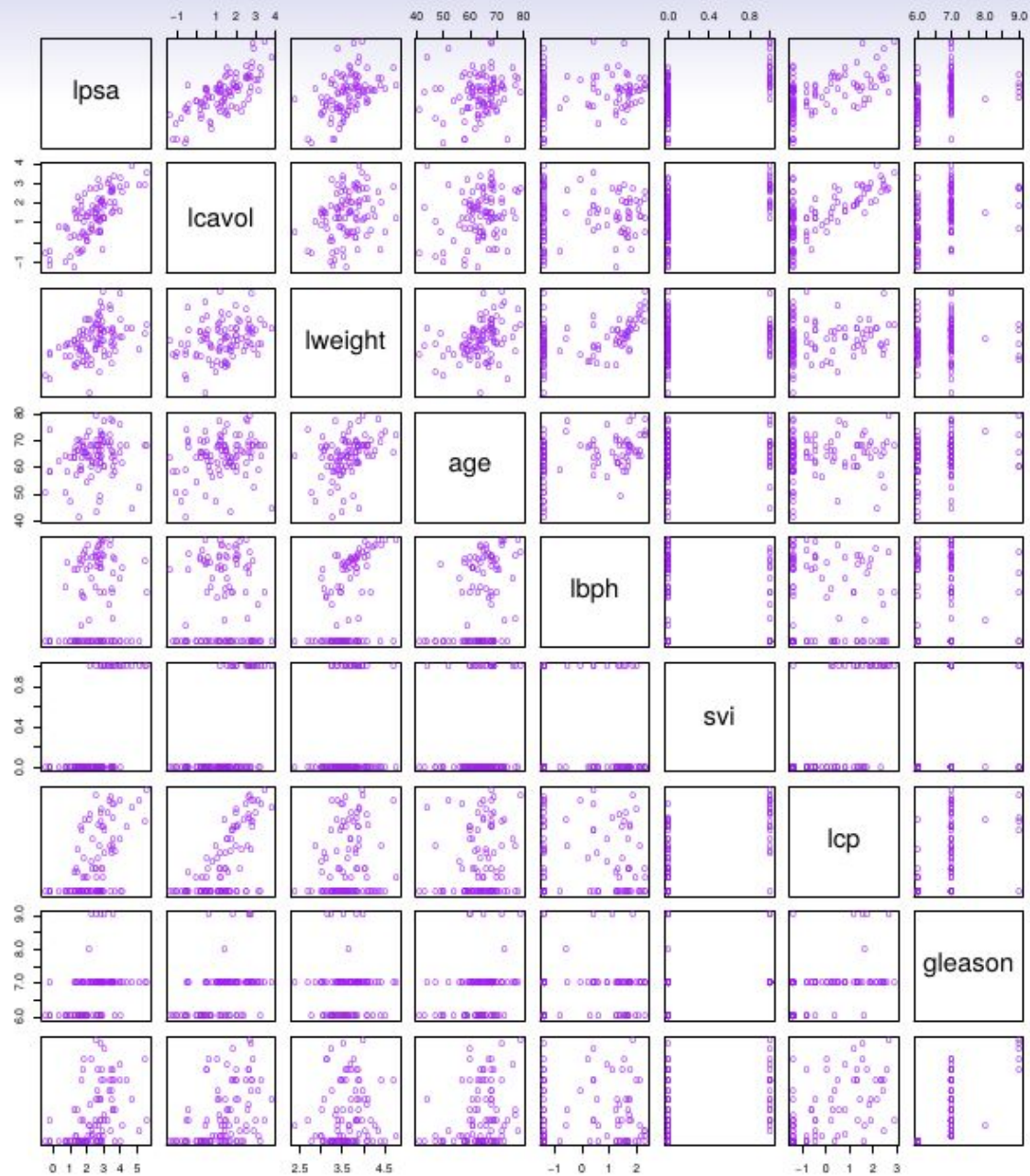
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Andrew Harnik/Associated Press

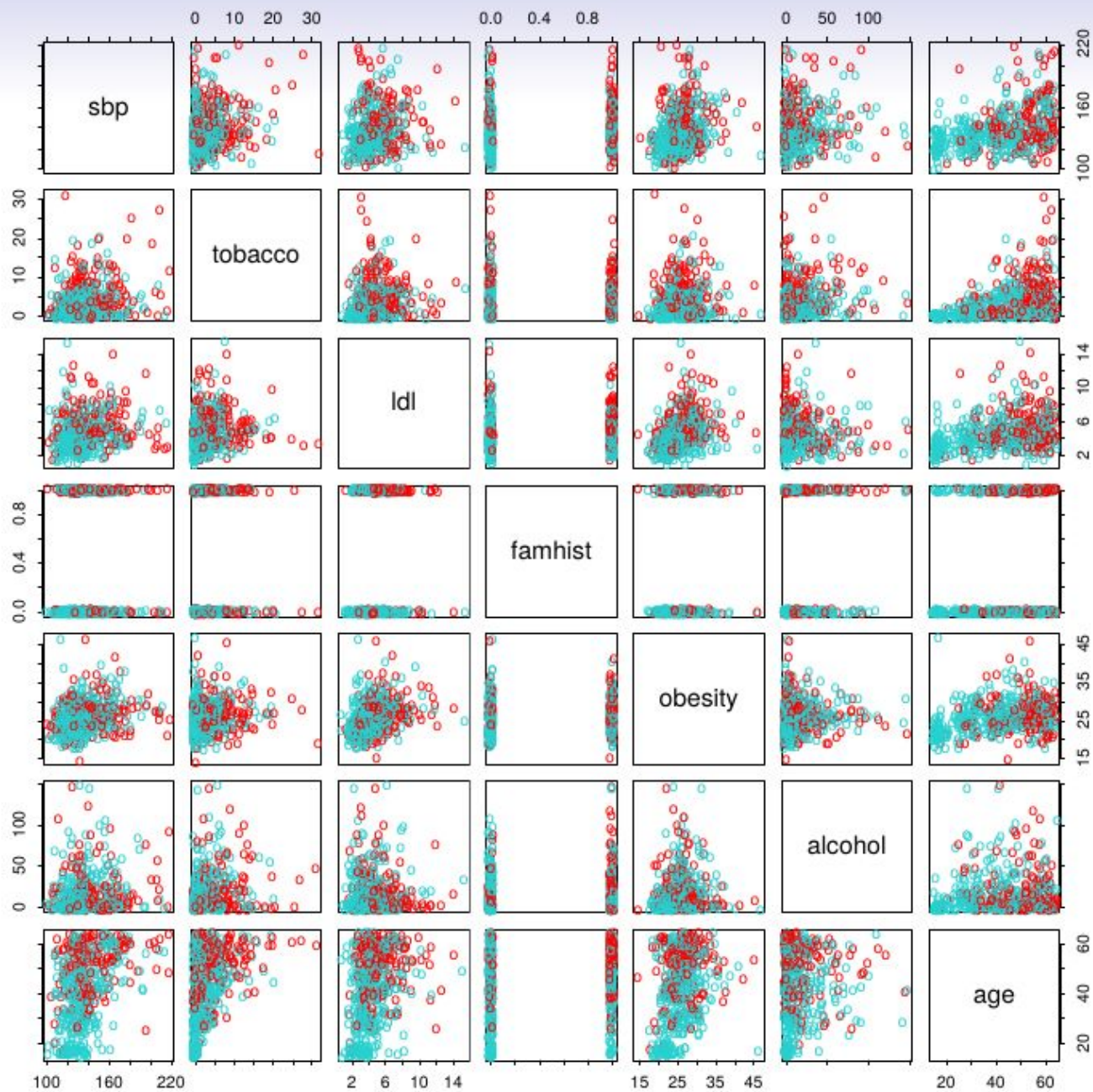
Statistical Learning Problems

- Identify the risk factors for prostate cancer.
- Classify a recorded phoneme based on a log-periodogram.
- Predict whether someone will have a heart attack on the basis of demographic, diet and clinical measurements.
- Customize an email spam detection system.
- Identify the numbers in a handwritten zip code.
- Classify a tissue sample into one of several cancer classes, based on a gene expression profile.
- Establish the relationship between salary and demographic variables in population survey data.
- Classify the pixels in a LANDSAT image, by usage.



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Spam Detection

- data from 4601 emails sent to an individual (named George, at HP labs, before 2000). Each is labeled as *spam* or *email*.
- goal: build a customized spam filter.
- input features: relative frequencies of 57 of the most commonly occurring words and punctuation marks in these email messages.

	george	you	hp	free	!	edu	remove
spam	0.00	2.26	0.02	0.52	0.51	0.01	0.28
email	1.27	1.27	0.90	0.07	0.11	0.29	0.01

*Average percentage of words or characters in an email message equal to the indicated word or character. We have chosen the words and characters showing the largest difference between **spam** and **email**.*

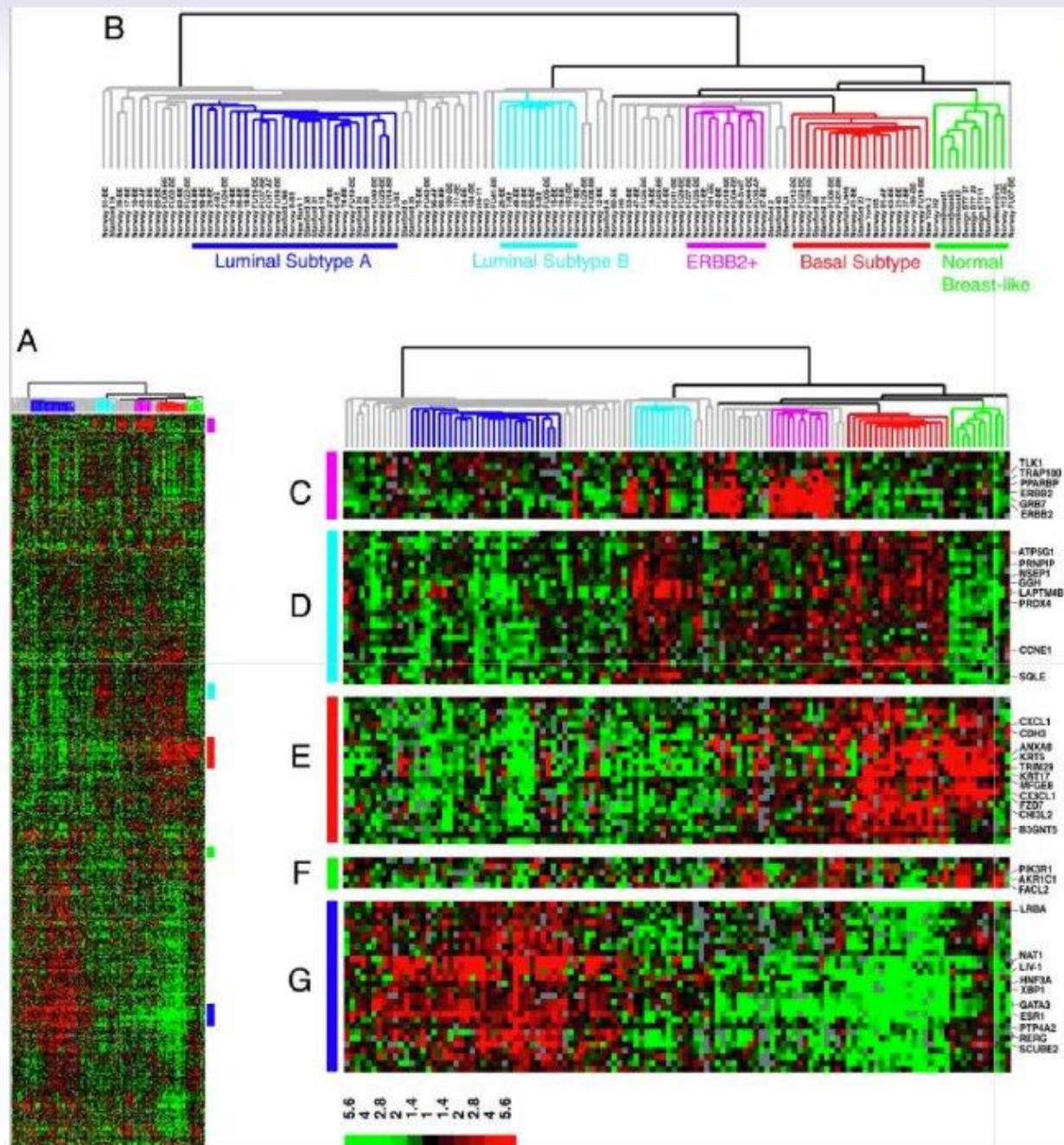
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0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
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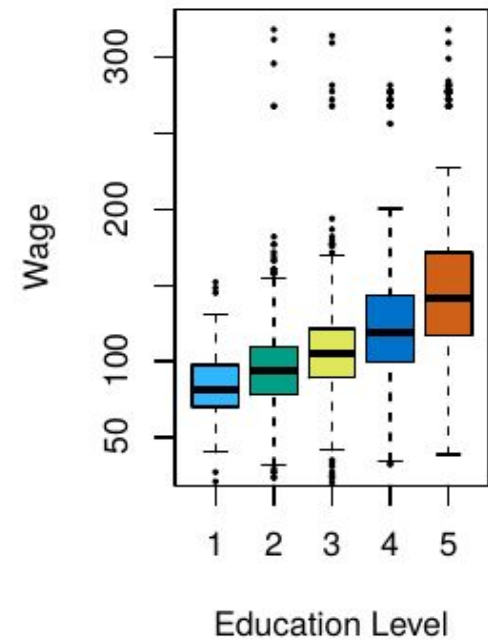
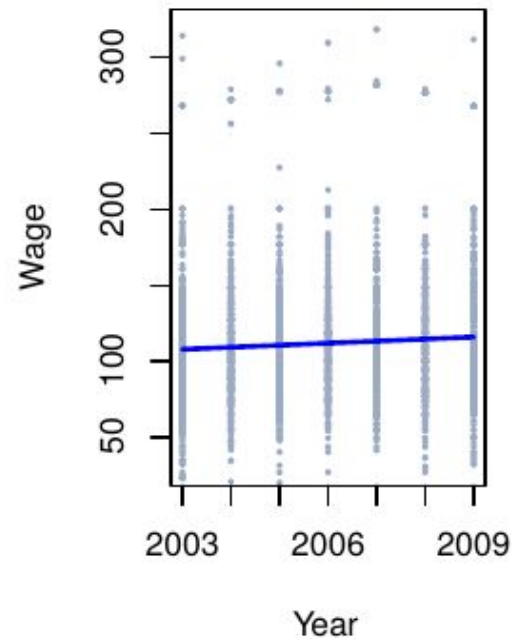
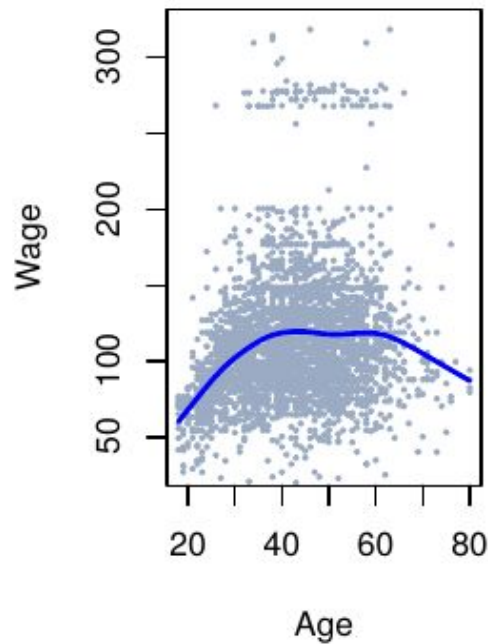
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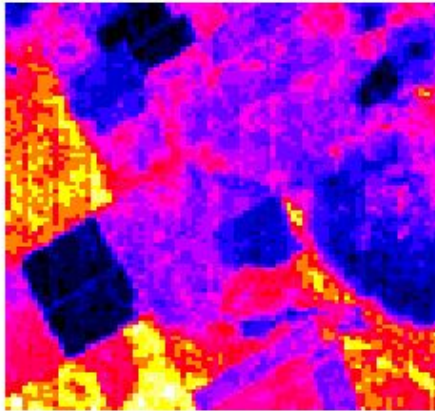


Income survey data for males from the central Atlantic region of the USA in 2009.

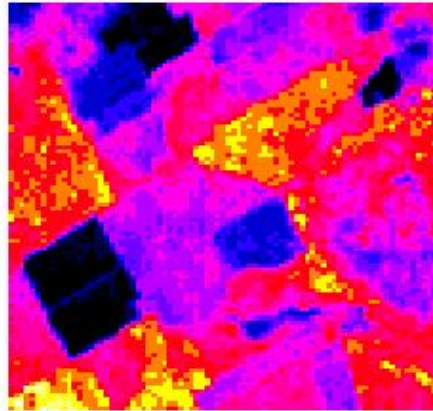
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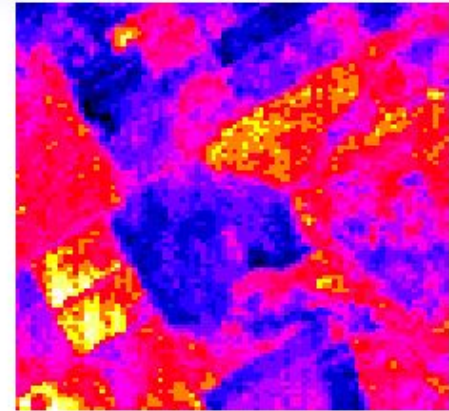
Spectral Band 1



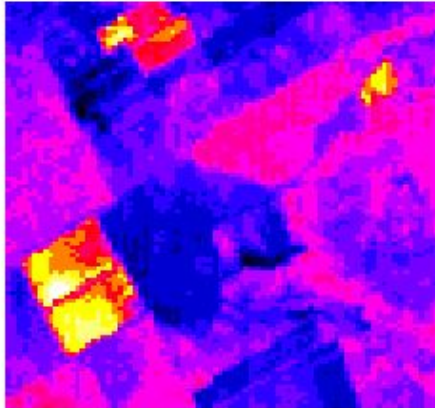
Spectral Band 2



Spectral Band 3



Spectral Band 4



Land Usage



Predicted Land Usage



$Usage \in \{\text{red soil, cotton, vegetation stubble, mixture, gray soil, damp gray soil}\}$

The Supervised Learning Problem

Starting point:

- Outcome measurement Y (also called dependent variable, response, target).
- Vector of p predictor measurements X (also called inputs, regressors, covariates, features, independent variables).
- In the *regression problem*, Y is quantitative (e.g price, blood pressure).
- In the *classification problem*, Y takes values in a finite, unordered set (survived/died, digit 0-9, cancer class of tissue sample).
- We have training data $(x_1, y_1), \dots, (x_N, y_N)$. These are observations (examples, instances) of these measurements.

Objectives

On the basis of the training data we would like to:

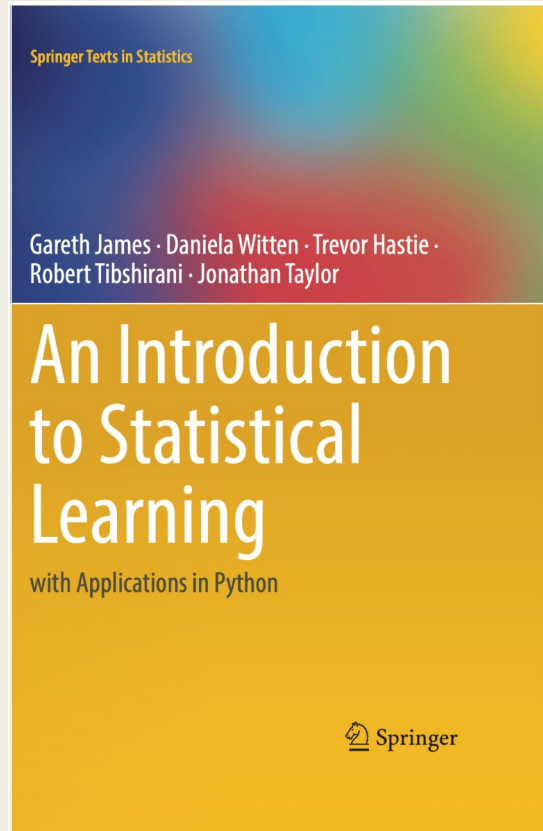
- Accurately predict unseen test cases.
- Understand which inputs affect the outcome, and how.
- Assess the quality of our predictions and inferences.

Unsupervised learning

- No outcome variable, just a set of predictors (features) measured on a set of samples.
- objective is more fuzzy — find groups of samples that behave similarly, find features that behave similarly, find linear combinations of features with the most variation.
- difficult to know how well you are doing.
- different from supervised learning, but can be useful as a pre-processing step for supervised learning.

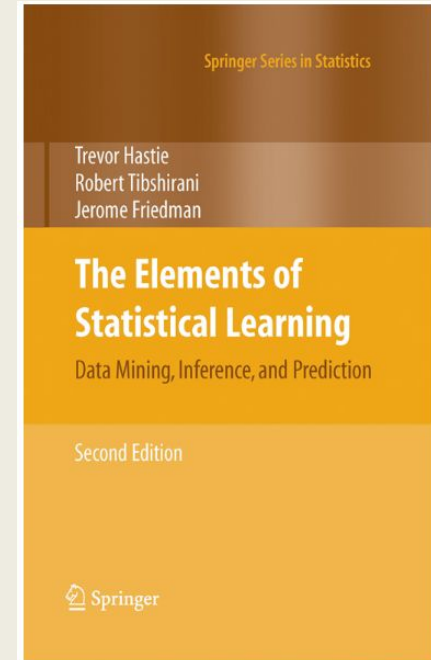
Brainstorm Examples of Supervised vs Unsupervised Learning

Course Texts



Covers almost everything in this course with a nice balance of rigor, mathematical simplicity and practicality.

Excellent,
free,
required.



More mathematical depth, more topics. Occasional optional readings.